

Ejer schwarz pick extremales

Ejercicio 1. Sean $\alpha, \beta \in \mathbb{D}$, definimos $\mathcal{A}_{\alpha, \beta} := \{f \in \mathcal{H}(\mathbb{D}) : f(\alpha) = \beta\}$. Demuestra que

- (a) $\mathcal{A}_{\alpha, \beta} \neq \emptyset$. (b) $\sup_{f \in \mathcal{A}_{\alpha, \beta}} |f'(\alpha)| = \frac{1 - |\beta|^2}{1 - |\alpha|^2}$.
- (c) Si $f \in \mathcal{A}_{\alpha, \beta}$ alcanza el supremo, entonces $f = \tau_\beta \circ \rho_\gamma \circ \tau_\alpha$ con $\gamma \in \mathbb{T}$ libre.

Solución:

(a) Basta con tomar $f = \tau_\beta \circ \tau_\alpha$, que es holomorfa y cumple que $f(\alpha) = \tau_\beta(0) = \beta$.

(b) Sea $f \in \mathcal{A}_{\alpha, \beta}$. Por el Teorema de Schwarz-Pick, tenemos que

$$|f'(\alpha)| \leq \frac{1 - |f(\alpha)|^2}{1 - |\alpha|^2} = \frac{1 - |\beta|^2}{1 - |\alpha|^2}.$$

Además, para $f = \tau_\beta \circ \tau_\alpha$, se satisface la igualdad:

$$|(\tau_\beta \circ \tau_\alpha)'(\alpha)| = |\tau_\beta'(0)| \cdot |\tau_\alpha'(\alpha)| \stackrel{\text{lem-involucion-disco-unidad-derivada}}{=} (|\beta|^2 - 1) \cdot \frac{1 - |\beta|^2}{|\alpha|^2 - 1} = \frac{1 - |\beta|^2}{1 - |\alpha|^2}.$$

$$\text{Luego } \sup_{f \in \mathcal{A}_{\alpha, \beta}} |f'(\alpha)| = \frac{1 - |\beta|^2}{1 - |\alpha|^2}.$$

- (c) Si $f \in \mathcal{A}_{\alpha, \beta}$ alcanza el supremo, entonces hay igualdad en la segunda desigualdad del teorema de Schwarz-Pick, luego $f \in \text{Aut}(\mathbb{D})$. Definimos $g := \tau_\beta \circ f \circ \tau_\alpha$. Entonces, $g \in \text{Aut}(\mathbb{D})$ y $g(0) = \tau_\beta(f(\alpha)) = \tau_\beta(\beta) = 0$. Por la obs-aut-disco-unidad-fija-origen-imp-rotacion $\exists \gamma \in \mathbb{T} : g = \rho_\gamma$. Por tanto, al despejar f , llegamos a $f = \tau_\beta \circ \rho_\gamma \circ \tau_\alpha$. ■

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